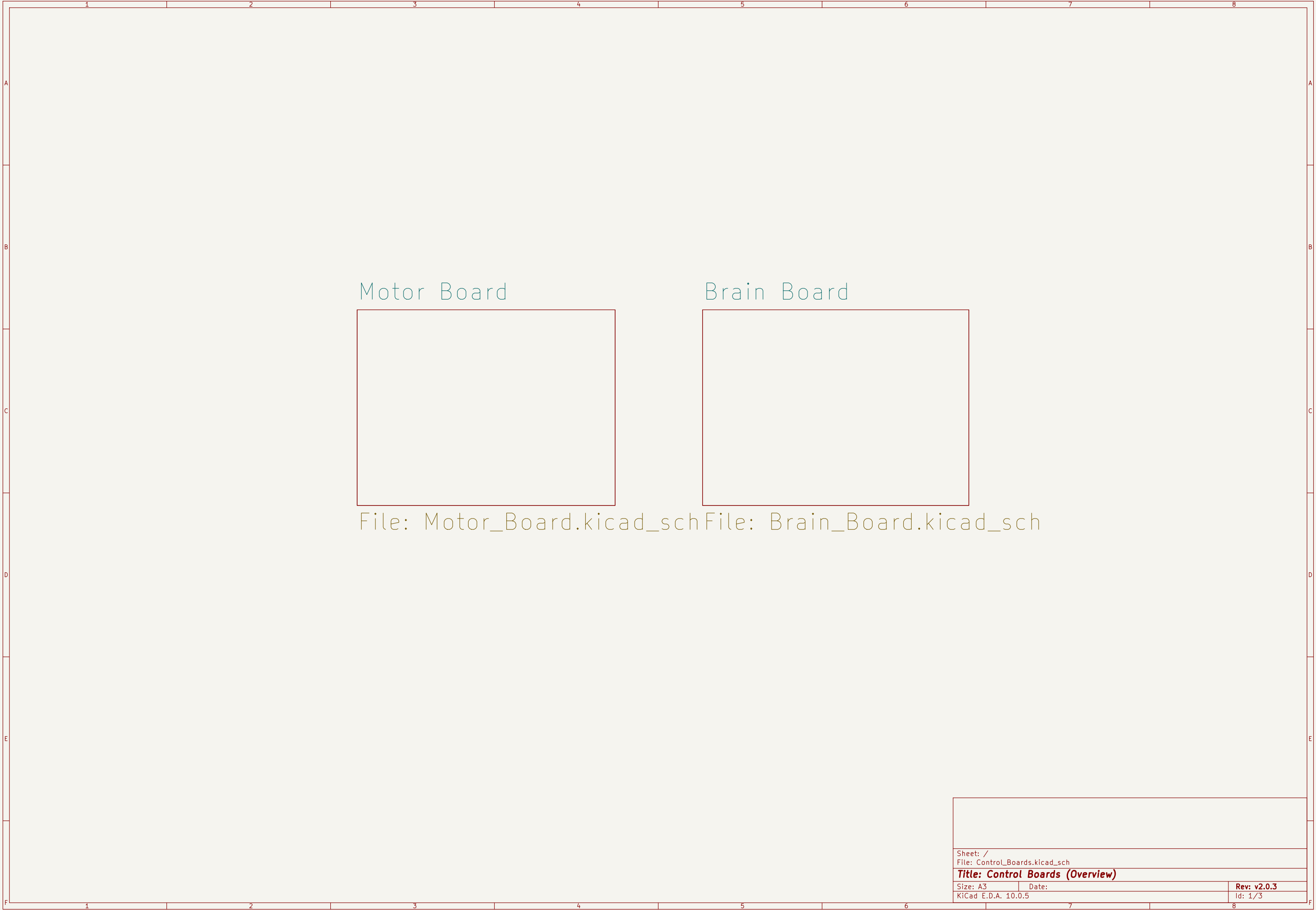
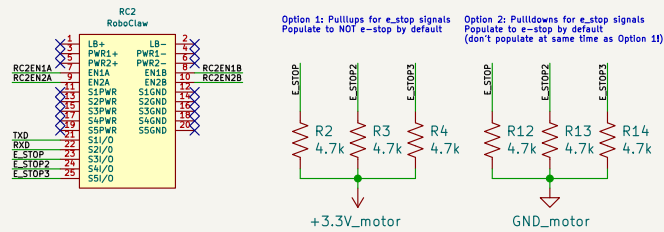
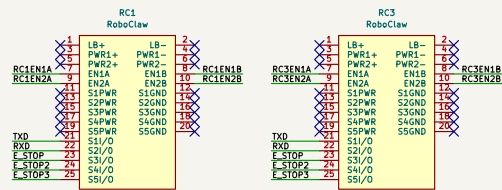
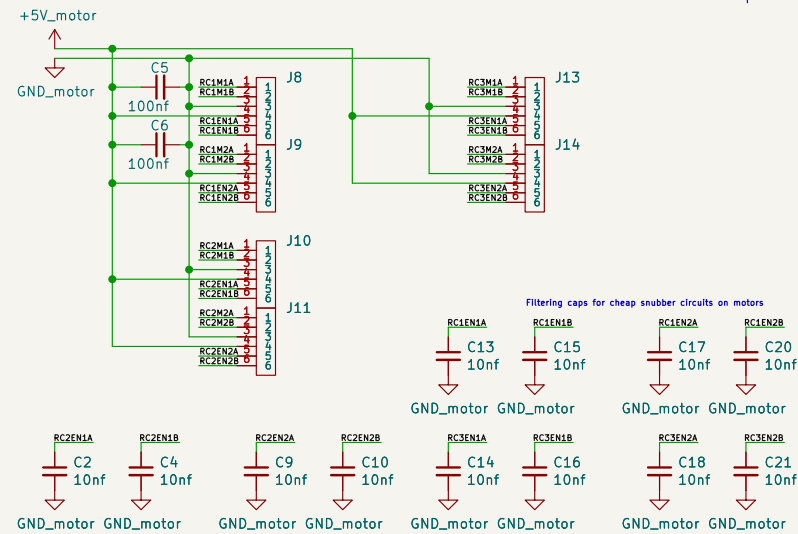




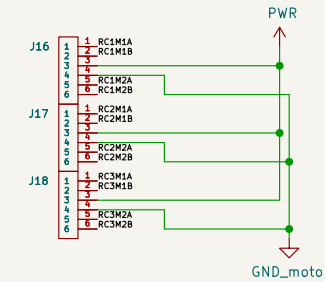
RoboClaw Pin Header



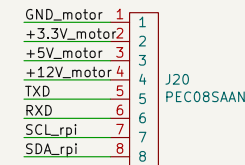
Roboclaw Motor Connection Output



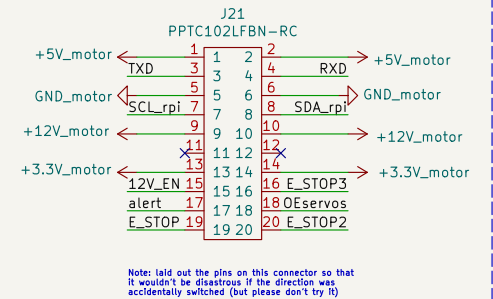
Roboclaw Power BUS



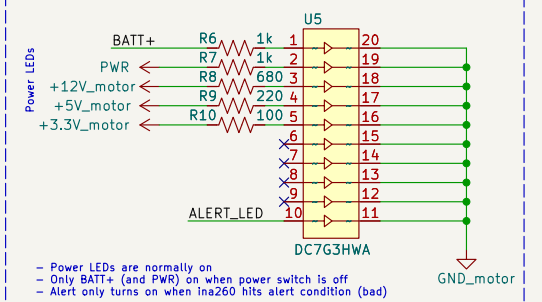
Test Pin Header



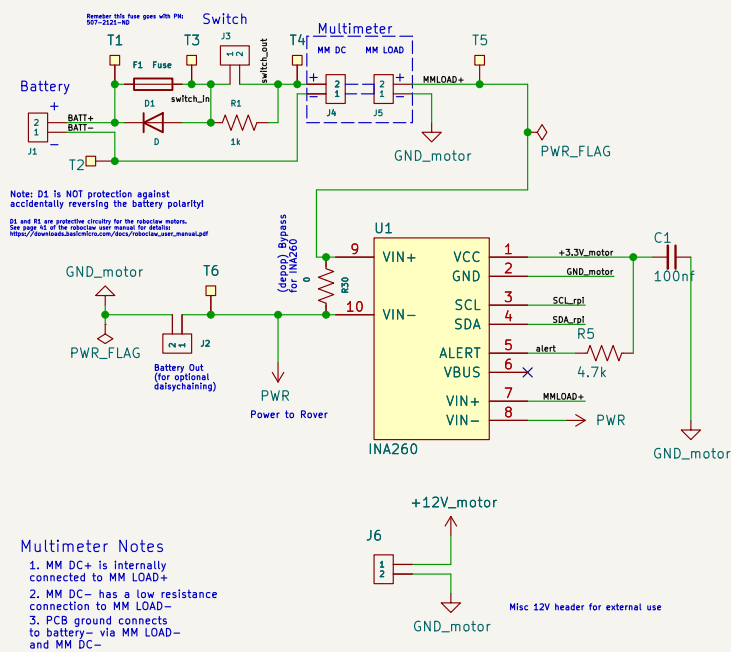
Inter-Board Connection



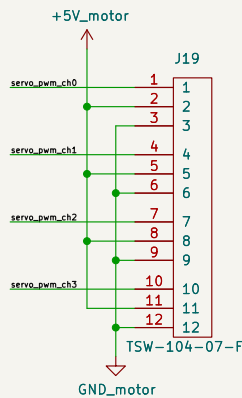
Motor LEDs



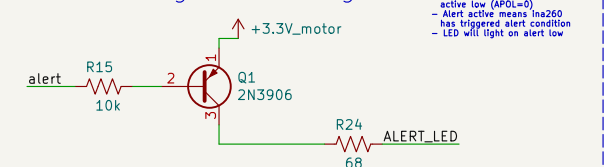
Power Distribution



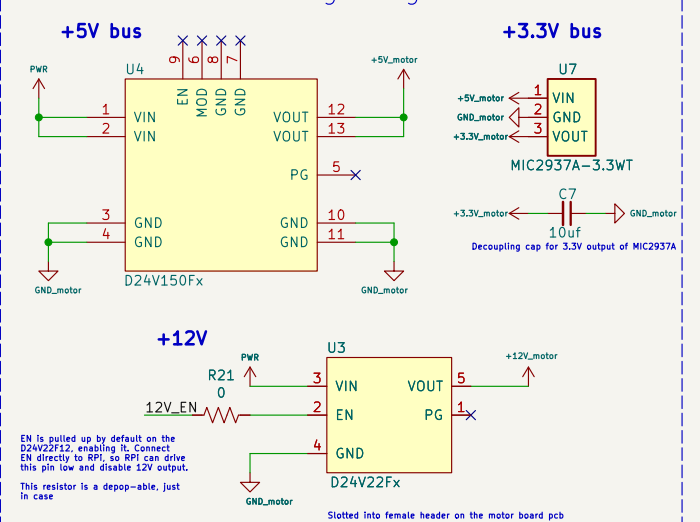
Corner Steering Servos Header



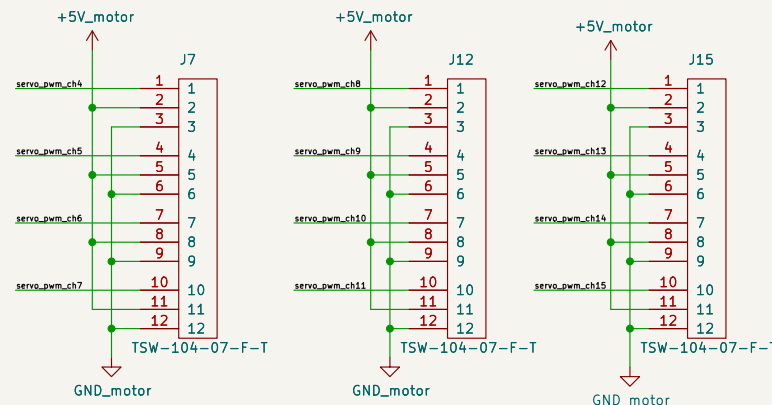
Alert Signal LED logic



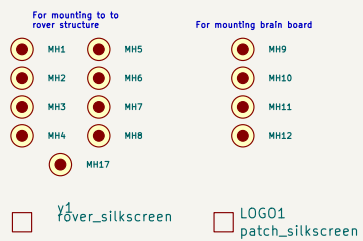
Voltage Regulators



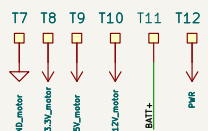
Extra PWM Outputs Headers



Miscellaneous



Power Test Points



Sheet: /Motor Board/
File: Motor_Board.kicad_sch

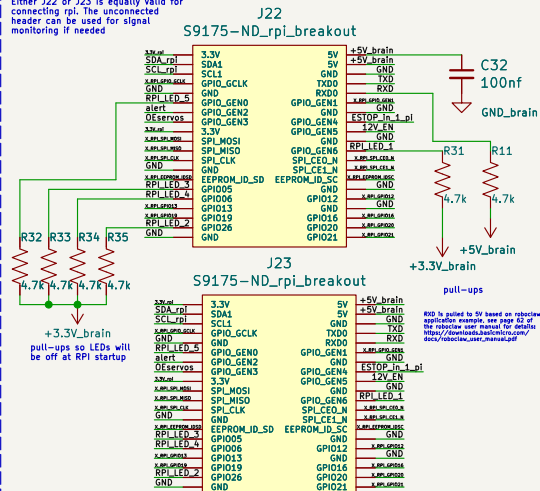
Title: Motor Board

Size: A3
Date:
KiCad E.D.A. 10.0.5

Rev: v2.0.3
Id: 2/3

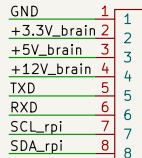
Raspberry Pi breakout

Either J22 or J23 is equally valid for connecting rpi. The unconnected header can be used for signal monitoring if needed

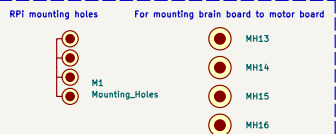


For GPIO availability, see worksheet here:
<https://www.notion.so/gpio-pins-Worksheet-9ebb23e9bbee4172a7b114ba388c0c59>

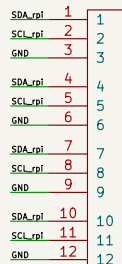
Test Pin Header



Voltage pins can be used to test for good connectivity brain <-> motor board

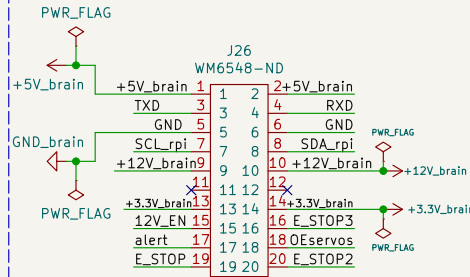


RPi I2C Header



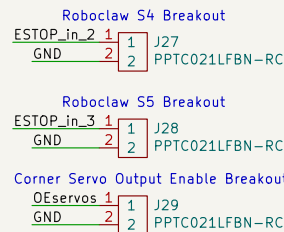
4 redundant I2C connections, for talking to arbitrary peripherals

Inter-Board Connection

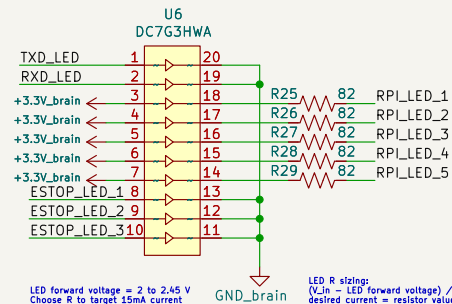


Note: laid out the pins on this connector so that it wouldn't be disastrous. If the direction was accidentally switched (but please don't try it)

Secondary ESTOP header



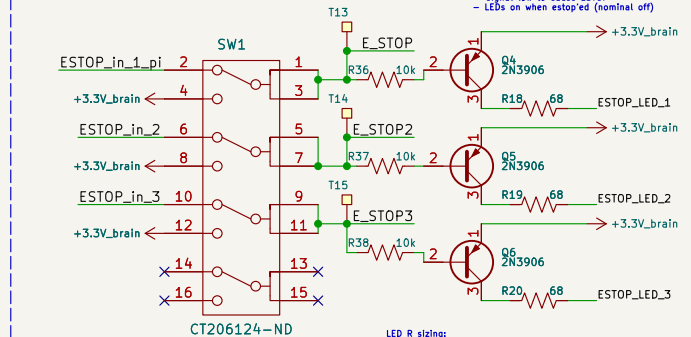
Brain LEDs



LED forward voltage = 2 to 2.45 V
 Choose R to target 15mA current

LED R sizing:
 $(V_{in} - LED \text{ forward voltage}) / \text{desired current} = \text{resistor value}$

ESTOP LED logic

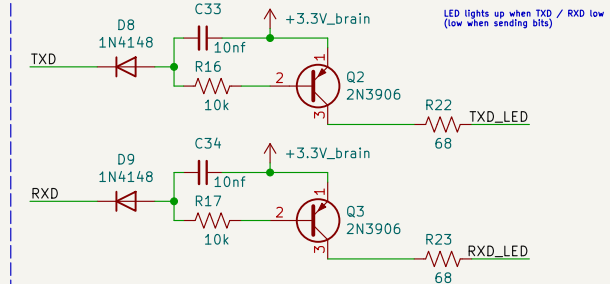


- Roboclaw ESTOPs active low (emergency stop when pulled low)
 - When enabled by SW1, pull an ESTOP_in signal low to cause ESTOP
 - LEDs on when estop'ed (nominal off)

Note that two adjacent odd numbered pins MUST be connected together, due to the implementation of the switch array

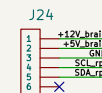
LED R sizing:
 $(V_{in} - \text{collector emitter saturation voltage} - LED \text{ forward voltage}) / \text{desired current} = \text{resistor value}$

Serial TXD/RXD LED logic



LED lights up when TXD / RXD low (low when sending bits)

Arduino Communication



Sheet: /Brain Board/
 File: Brain_Board.kicad_sch

Title: Brain Board

Size: A4
 KiCad E.D.A. 10.0.5

Date:

Rev: v2.0.3
 Id: 3/3